Lab 7.2 - Implement Inter-VLAN Routing

Important for Online Workshop Students Only:

You must use Packet Tracer v8.2.2 or higher to complete this lab.

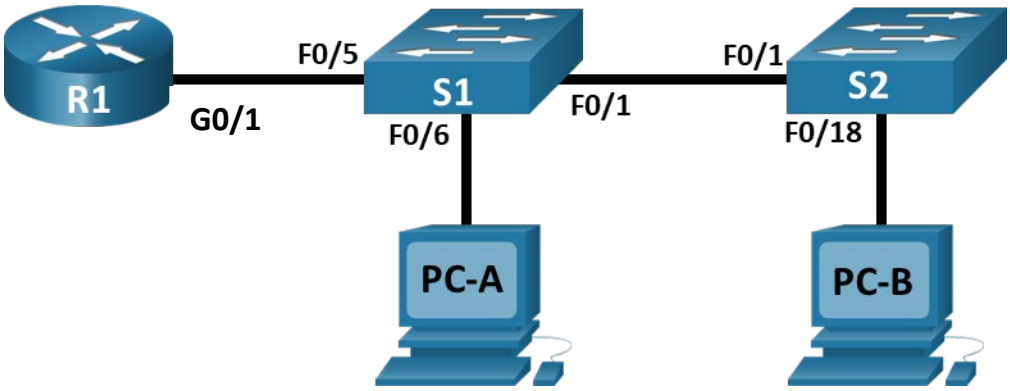
Please rename your Packet Tracer file to FirstName_LastName_sNumber_Lab7.2.pka

Use the 2911 router and 2960 switch

Scoring

No.		Score
1	PCs	8 each
2	S1	45
3	S2	36
4	R1	39

Topology



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Addressing Table

Device	Interface	IP Address / Prefix	Default Gateway
R1	G0/1.10	192.168.10.1 /24	N/A
		2001:DB8:ACAD:A::1 /64	
	G0/1.20	192.168.20.1 /24	
		2001:DB8:ACAD:B::1 /64	
	G0/1.30	192.168.30.1 /24	
		2001:DB8:ACAD:C::1 /64	
	G0/1.1000	N/A	
S1	VLAN 10	192.168.10.11 /24	192.168.10.1
S2	VLAN 10	192.168.10.12 /24	192.168.10.1
PC-A	NIC	192.168.20.3 /24	192.168.20.1
		2001:DB8:ACAD:B::3 /64	2001:DB8:ACAD:B::1 /64
PC-B	NIC	192.168.30.3 /24	192.168.30.1
		2001:DB8:ACAD:C::3 /64	2001:DB8:ACAD:C::1 /64

VLAN Table

VLAN	Name	Interface Assigned
10	Management	S1: VLAN 10 S2: VLAN 10
20	Sales	S1: F0/6
30	Operations	S2: F0/18
999	Parking_Lot	S1: F0/2-4, F0/7-24, G0/1-2 S2: F0/2-17, F0/19-24, G0/1-2
1000	Native	N/A

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Create VLANs and Assign Switch Ports

Part 3: Configure an 802.1Q Trunk between the Switches

Part 4: Configure Inter-VLAN Routing on the Router

Part 5: Verify Inter-VLAN Routing is working

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Background / Scenario

Modern switches use virtual local-area networks (VLANs) to improve network performance by separating large Layer 2 broadcast domains into smaller ones. VLANs can also be used as a security measure by separating sensitive data traffic from the rest of the network. In general, VLANs make it easier to design a network to support the goals of an organization. Communication between VLANs requires a device operating at Layer 3 of the OSI model. Adding an inter-VLAN router allows the organization to segregate and separate broadcast domains while simultaneously allowing them to communicate with each other.

VLAN trunks are used to span VLANs across multiple devices. Trunks allow the traffic from multiple VLANs to travel over a single link, while keeping the VLAN identification and segmentation intact. A particular kind of inter-VLAN routing, called “Router-on-a-Stick”, uses a trunk from the router to the switch to enable all VLANs to pass to the router.

In this lab, you will create VLANs on both switches in the topology, assign VLANs to switch access ports, verify that VLANs are working as expected, create VLAN trunks between the two switches and between S1 and R1, and configure Inter-VLAN routing on R1 to allow hosts in different VLANs to communicate, regardless of which subnet the host resides.

Note: For students in on-campus workshops only: Make sure that the switch has been erased and has no startup configurations. Please refer to Appendix A

Note: For students in online workshops only: In Packet Tracer, you must use the 2911 router and the 2960 switch

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings on the PC hosts and switches.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for the router.

a. Assign a device name to the router.

b. Disable DNS lookup to prevent the router from attempting to translate incorrectly entered commands as though they were host names.

c. Assign **class** as the privileged EXEC encrypted password.

d. Assign **cisco** as the console password and enable login.

e. Assign **cisco** as the vty password and enable login.

f. Configure logging synchronous for the console line and vty lines.

g. Encrypt the plaintext passwords.

h. Create a banner that warns anyone accessing the device that unauthorized access is prohibited with the following text:

Unauthorised access will be prosecuted.

i. Configure the router for SSH access using RSA and 1024 bits. Use privilege level **15** username **admin** and password **cisco**.

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Step 3: Configure basic settings for each switch.

a. Assign a device name to the switch.

b. Disable DNS lookup to prevent the switch from attempting to translate incorrectly entered commands as though they were host names.

c. Assign **class** as the privileged EXEC encrypted password.

d. Assign **cisco** as the console password and enable login.

e. Assign **cisco** as the vty password and enable login.

f. Configure logging synchronous for the console line and vty lines.

g. Encrypt the plaintext passwords.

h. Create a banner that warns anyone accessing the device that unauthorized access is prohibited with the following text:

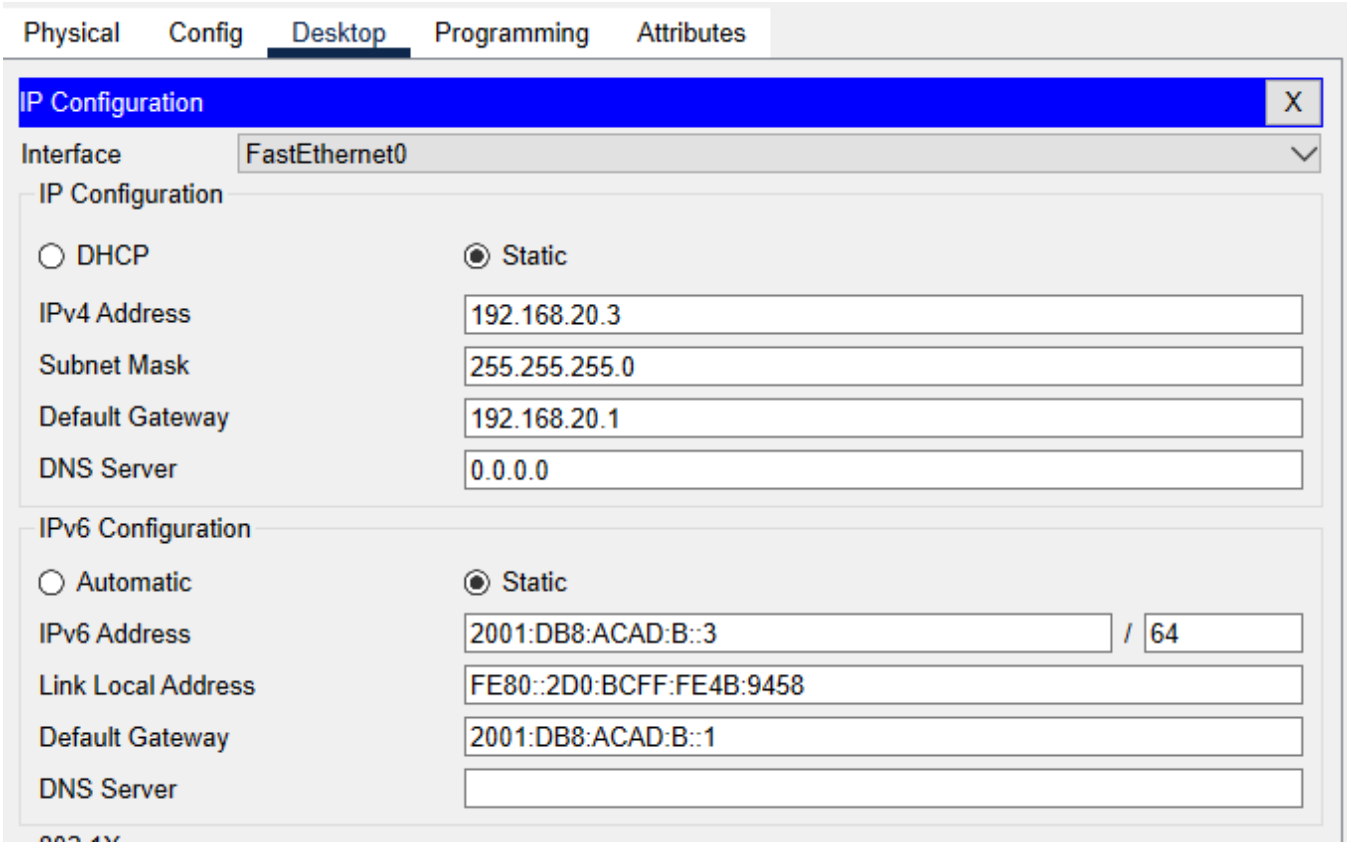
Unauthorised access will be prosecuted.

i. Configure the switch for SSH access using RSA and 1024 bits. Use privilege level **15** username **admin** and password **cisco**.

Step 4: Configure PC hosts.

Refer to the Addressing Table for PC host address information.

Example configuration for PC-A for Packet Tracer version: Note the default gateway configuration.



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Part 2: Create VLANs and Assign Switch Ports

In Part 2, you will create VLANs as specified in the table above on both switches. You will then assign the VLANs to the appropriate interface and verify your configuration settings. Complete the following tasks on each switch.

Step 1: Create VLANs on both switches.

a. Create and name the required VLANs on each switch from the table above.

b. Configure the management interface and default gateway on each switch using the IP address information in the Addressing Table.

To configure the default IPv4 gateway on a switch use the following command:

S1:

```
S1(config)# ip default-gateway 192.168.10.1
```

S2:

```
S2(config)# ip default-gateway 192.168.10.1
```

Note, we will not be configuring IPv6 on the switches.

c. Assign all unused ports on the switch to the Parking_Lot VLAN, configure them for static access mode, and administratively deactivate them.

Note: The **interface range** command is helpful to accomplish this task with as few commands as necessary.

Step 2: Assign VLANs to the correct switch interfaces.

a. Assign used ports to the appropriate VLAN (specified in the VLAN table above) and configure them for static access mode.

b. Issue the **show vlan brief** command and verify that the VLANs are assigned to the correct interfaces.

Part 3: Configure an 802.1Q Trunk Between the Switches

In Part 3, you will manually configure interfaces F0/1 and F0/5 as a trunk.

Step 1: Manually configure trunk interface F0/1 on switch S1 and S2.

a. Configure static trunking on interface F0/1 for both switches.

b. Set the native VLAN to 1000 on both switches.

c. Specify that VLANs 10, 20, 30, and 1000 are allowed to cross the trunk.

d. Issue the **show interfaces trunk** command to verify trunking ports, the Native VLAN and allowed VLANs across the trunk.

Step 2: Manually configure S1's trunk interface F0/5

a. Configure S1's interface F0/5 with the same trunk parameters as F0/1. This is the trunk to the router.

b. Issue the **show interfaces trunk** command to verify trunking.

Why does F0/5 not appear in the list of trunks?

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Part 4: Configure Inter-VLAN Routing on the Router

Step 1: Configure the router.

a. Within global configuration mode, enable IPv6 routing on R1.

```
R1(config)# ipv6 unicast-routing
```

b. Configure sub-interfaces for each VLAN as specified in the IP addressing table. All sub-interfaces use 802.1Q encapsulation. Ensure the sub-interface for the native VLAN does not have an IP address assigned. Include a description for each sub-interface.

VLAN 10:

```
R1(config)# interface g0/1.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config-subif)# ipv6 address 2001:DB8:ACAD:A::1/64
R1(config-subif)# description "VLAN 10"
R1(config-subif)# exit
```

VLAN 20:

```
R1(config)# interface g0/1.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
R1(config-subif)# ipv6 address 2001:DB8:ACAD:B::1/64
R1(config-subif)# description "VLAN 20"
R1(config-subif)# exit
```

VLAN 30:

```
R1(config)# interface g0/1.30
R1(config-subif)# encapsulation dot1Q 30
R1(config-subif)# ip address 192.168.30.1 255.255.255.0
R1(config-subif)# ipv6 address 2001:DB8:ACAD:C::1/64
R1(config-subif)# description "VLAN 30"
R1(config-subif)# exit
```

VLAN 1000 (native):

```
R1(config)# interface g0/1.1000
R1(config-subif)# encapsulation dot1Q 1000 native
R1(config-subif)# description "native VLAN 1000"
R1(config-subif)# exit
```

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c. Activate interface G0/1 on the router.

```
R1(config)# interface g0/1
R1(config)# no shutdown
R1(config)# exit
```

d. Use the **show ip interface brief** and **show ipv6 interface brief** commands to verify the sub-interfaces are operational.

Part 5: Verify Inter-VLAN Routing is working

Step 1: Complete the following tests from PC-A. All should be successful.

a. For both IPv4 and IPv6, ping from PC-A to its default gateway.

b. For both IPv4 and IPv6, ping from PC-A to PC-B.

c. For IPv4, ping from PC-A to S2.

Step 2: Complete the following test from PC-B

From the Command Prompt window on PC-B, issue the **tracert** command to the IPv4 address of PC-A.

What intermediate IP addresses are shown in the results?

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Appendix A: Initializing and Reloading a Router and Switch

Step 1: Initialize and reload the router.

a. Console into the router and enable privileged EXEC mode.

```
Router> enable
Router#
```

b. Type the **erase startup-config** command to remove the startup configuration from NVRAM.

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
Router#
```

c. Issue the **reload** command to remove an old configuration from memory. When prompted to **Proceed with reload**, press Enter to confirm the reload. (Pressing any other key aborts the reload.)

```
Router# reload
Proceed with reload? [confirm]
*Nov 29 18:28:09.923: %SYS-5-RELOAD: Reload requested by console. Reload Reason:
Reload Command.
```

Note: You may be prompted to save the running configuration prior to reloading the router. Type **no** and press Enter.

```
System configuration has been modified. Save? [yes/no]: no
```

d. After the router reloads, you are prompted to enter the initial configuration dialog. Enter **no** and press Enter.

```
Would you like to enter the initial configuration dialog? [yes/no]: no
```

e. You are prompted to terminate autoinstall. Type **yes** and then press Enter.

```
Would you like to terminate autoinstall? [yes]: yes
```

Step 2: Initialize and reload the switch.

a. Console into the switch and enter privileged EXEC mode.

```
Switch> enable
Switch#
```

b. Use the **show flash** command to determine if any VLANs have been created on the switch.

```
Switch# show flash
Directory of flash:/

    2  -rwx        1919   Mar 1 1993 00:06:33 +00:00  private-config.text
    3  -rwx        1632   Mar 1 1993 00:06:33 +00:00  config.text
    4  -rwx       13336   Mar 1 1993 00:06:33 +00:00  multiple-fs
    5  -rwx    11607161   Mar 1 1993 02:37:06 +00:00  c2960-lanbasek9-mz.150-2.SE.bin
    6  -rwx         616   Mar 1 1993 00:07:13 +00:00  vlan.dat

32514048 bytes total (20886528 bytes free)
Switch#
```

c. If the **vlan.dat** file was found in flash, then delete this file.

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```
Switch# delete vlan.dat
Delete filename [vlan.dat]?
```

d. You are prompted to verify the filename. At this point, you can change the filename or just press Enter if you have entered the name correctly.

e. You are prompted to confirm deleting this file. Press Enter to confirm deletion. (Pressing any other key aborts the deletion.)

```
Delete flash:/vlan.dat? [confirm]
Switch#
```

f. Use the **erase startup-config** command to erase the startup configuration file from NVRAM. You are prompted to confirm removing the configuration file. Press Enter to confirm to erase this file. (Pressing any other key aborts the operation.)

```
Switch# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
Switch#
```

g. Reload the switch to remove any old configuration information from memory. You are prompted to confirm reloading the switch. Press Enter to proceed with the reload. (Pressing any other key aborts the reload.)

```
Switch# reload
Proceed with reload? [confirm]
```

Note: You may be prompted to save the running configuration prior to reloading the switch. Type **no** and press Enter.

```
System configuration has been modified. Save? [yes/no]: no
```

h. After the switch reloads, you should be prompted to enter the initial configuration dialog. Type **no** and press Enter.

```
Would you like to enter the initial configuration dialog? [yes/no]: no
Switch>
```

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